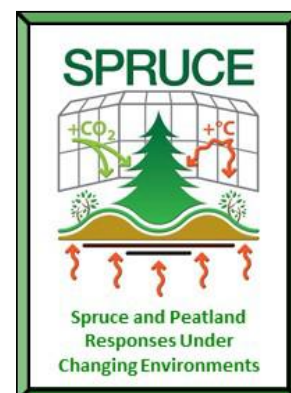


SPRUCE: Quantitative PCR (qPCR) of Microbial Gene Copy Numbers, 2021-2022

Record_id: spruce.110

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1. Dataset Overview:

Summary:

This dataset provides the results for quantitative polymerase chain reaction (qPCR) of peat samples collected from ambient and experimental plots in the Spruce and Peatland Responses Under Climatic and Environmental Change (SPRUCE) experiment site in June and August of 2021, and June of 2022. SPRUCE is located within the Marcell Experimental Forest in northern Minnesota, USA. The dataset includes bacterial, archaeal, fungal gene copy numbers, along with corresponding logarithmic values, at 11 depth increments of two-meter deep peat cores taken from 12 sampling sites locations inside SPRUCE plots (10 chambered and 2 ambient plots). The sampling, sample prep and analysis followed standard methods outlined in prior publications (Wilson et al. 2016; Kluber et al. 2020) except that a higher yielding Omega Bio-Tek Mag-Bind Environmental DNA 96 Kit was used for extractions and DNA was quantified using Qubit dsDNA High Sensitivity Assay Kit. qPCR subsamples of peat cores from the SPRUCE plots characterize changes in the abundance and composition of microbial communities of peat seasonally showing how composition varies under multiple levels of experimental peat warming and atmospheric CO₂ concentrations.

This dataset contains one data file in comma-separated values (.csv) format. Additional metadata are provided: one data dictionary and a file-level metadata file in comma-separated values (.csv) format and a user guide in PDF (*.pdf) format.

On 2026-07-07 this dataset was updated to add three columns to the data file: 'Fungal_copy_dry', 'Log_fungal_copy_dry', 'Fungal_copy_wet'. No previously released data values were altered. Additionally, the abstract, data dictionary, and user guide were updated, and a file-level metadata file was added.

Data Citation:

Cite this dataset as follows:

Roth, S, Z Buell, B Kristy, J Philips, S Ottinger, K Lowe, MA Mayes, and CW Schadt. 2024. **SPRUCE Quantitative PCR of Microbial Gene Copy Numbers, 2021-2022**. ORNL Terrestrial Ecosystem Science SFA, ESS-DIVE repository. Dataset. <https://doi.org/10.25581/spruce.110/1998878>

Related Datasets:

Buell, Z.W., J. Dabbs, J.M. Steinweg, J.R. Phillips, L.A. Kluber, Z.K. Yang, R.M. Miller, J.L.M. Gutknecht, C.W. Schadt, and M.A. Mayes. 2024. **Interrelationships among methods of estimating microbial biomass across multiple soil orders and biomes: Supporting data**. Oak Ridge National Laboratory, U.S. Department of Energy, Oak Ridge, Tennessee, U.S.A. <https://doi.org/10.25581/ornlsfa.031/2274949>.

- Data used in the present dataset are summarized in this dataset.

Buell, Z., J. Philips, S. Ottinger, K. Lowe, C.W. Schadt, & M.A. Mayes. 2024. **Chloroform Fumigation Extraction for Microbial Biomass and Dissolved Organic Carbon from SPRUCE, 2021-2022**. Oak Ridge National Laboratory, TES SFA, U.S. Department of Energy, Oak Ridge, Tennessee, U.S.A. <https://doi.org/10.25581/spruce.109/1998876>.

Buell, Z., M. Felice, J. Philips, S. Ottinger, K. Lowe, & J.L.M. Gutknecht. 2024. **SPRUCE Phospholipid Fatty Acid (PLFA) Abundances, August 2021-June 2022**. Oak Ridge National Laboratory, TES SFA, U.S. Department of Energy, Oak Ridge, Tennessee, U.S.A. <https://doi.org/10.25581/spruce.112/1998897>.

Hanson, P.J., J.S. Riggs, W.R. Nettles, M.B. Krassovski, and L.A. Hook. 2016. **SPRUCE Whole Ecosystems Warming (WEW) Environmental Data Beginning August 2015**. Oak Ridge National Laboratory, TES SFA, U.S. Department of Energy, Oak Ridge, Tennessee, U.S.A. <https://doi.org/10.3334/CDIAC/spruce.032>.

Dataset Release History

Release Date	Release Notes	Data Range
2026-07-07	Second Public Release Added three columns to the data file: 'Fungal_copy_dry', 'Log_fungal_copy_dry', 'Fungal_copy_wet'. Updated the abstract, added a file-level metadata file, and made minor edits to the user	2021-06-23 to 2022-06-21

	guide. No previously released data values have been altered.	
2024-01-25	Initial Public Release Initial release of 1 data file	2021-06-23 to 2022-06-21

2. Data Characteristics:

This dataset contains one comma-separated values (*.csv) file:

- **SPRUCE_qPCR_copy_numbers_2021.csv:** Includes bacterial and archaeal gene copy numbers, along with corresponding logarithmic values, at 11 depth increments of two-meter deep peat cores taken from 12 sampling sites locations inside SPRUCE plots

Spatial Coverage

All measurements were made at the 8.1-ha S1 bog forest site in northern Minnesota, 40 km north of Grand Rapids, in the USDA Forest Service Marcell Experimental Forest (MEF). The following coordinates are the central location of the S1 bog.

Site boundaries: Latitude and longitude given in decimal degrees.

Site	Westernmost Longitude	Easternmost Longitude	Northernmost Latitude	Southernmost Latitude	Elevation (meters amsl)	Geodetic Datum
S1 Bog, Marcell Experimental Forest	-93.45399	-93.45256	47.50656	47.50470	418	WGS84

Plot characteristics: Plots where observations were made are in bold. Additional plot characteristics can be found in the [SPRUCE Plot Characteristics](#) table and [Tree and Equipment Positions](#) document.

Plot	Experimental Plot	With Enclosure	Temperature Treatment (degrees C)	Carbon Dioxide Treatment (elevated = 500 ppm above ambient)
2	No	No	Ambient	Ambient
4	Yes	Yes	+4.5	Elevated
5	No	No	Ambient	Ambient

6	Yes	Yes	+0 (blowers only)	Ambient
7	Yes	No	Ambient	Ambient
8	Yes	Yes	+6.75	Ambient
9	No	No	Ambient	Ambient
10	Yes	Yes	+9	Elevated
11	Yes	Yes	+2.25	Elevated
13	Yes	Yes	+4.5	Ambient
14	No	No	Ambient	Ambient
15	No	No	Ambient	Ambient
16	Yes	Yes	+6.75	Elevated
17	Yes	Yes	+9	Ambient
19	Yes	Yes	+0 (blowers only)	Elevated
20	Yes	Yes	+2.25	Ambient
21	Yes	No	Ambient	Ambient

Temporal Coverage

This dataset includes samples collected on 3 dates: 2021-06-23, 2021-08-24, 2022-06-21

Data File Description

These data are considered at **Quality Level 1**. Level 1 indicates an internally consistent data product that has been subjected to quality checks and data management procedures.

The provided data file is in comma-separated values (*.csv) format:

- **SPRUCE_qPCR_copy_numbers_2021.csv**: Contains quantitative polymerase chain reaction (qPCR) results from peat samples collected from SPRUCE in June and August of 2021, and June of 2022. The data file includes bacterial, archaeal, fungal gene copy numbers, along with corresponding logarithmic values.

Missing numeric data are indicated by -9999.

Data Dictionary for *SPRUCE_qPCR_copy_numbers_2021.csv*

Column Name	Units	Description
Sample_ID		Month_Year_Location_Depth-Depth
Sample		Location_Depth-Depth
Sample_date	YYYY-MM-DD	Date that sample was collected
Year	YYYY	Year in which sample was collected
Month		Month in which sample was collected
Depth_range	cm	Increment at which sample was taken from soil profile

Depth_avg	cm	Average depth of increment at which sample was taken
Plot		Location (experimental plot) which sample was taken from
Temp_experimental	Degrees C	Target temperature of plot above ambient levels.
Temp_annual_avg	Degrees C	Average annual temperature inside experimental plot
CO2_treatment		Plot temperature treatment. E = Elevated ; A = Ambient
Bacteria_copy_dry	gene copy number g-1 dry soil	Bacterial copy number per gram of dry soil
Log_bacteria_copy_dry	Log gene copy number g-1 dry soil	Log transformed bacterial copy number per gram of dry soil
Archaea_copy_dry	gene copy number g-1 dry soil	Archaeal copy number per gram of dry soil
Log_archaea_copy_dry	Log gene copy number g-1 dry soil	Log transformed archaeal copy number per gram of dry soil
Bacteria_copy_wet	gene copy number g-1 wet soil	Bacterial copy number per gram of field-wet soil
Archaea_copy_wet	gene copy number g-1 wet soil	Archaeal copy number per gram of field-wet soil
Fungal_copy_dry	gene copy number g-1 dry soil	Fungal copy number per gram of dry soil
Log_fungal_copy_dry	Log gene copy number g-1 dry soil	Log transformed Fungal copy number per gram of dry soil
Fungal_copy_wet	gene copy number g-1 wet soil	Fungal copy number per gram of field-wet soil

3 Applications and Derivation

qPCR subsamples of peat cores from the SPRUCE experimental plots are used to characterize changes in the abundance and composition of microbial communities of peat seasonally and how this composition varies under multiple levels of experimental peat warming and atmospheric CO₂ concentrations.

4. Quality Assessment:

These data are considered at **Quality Level 1**. Level 1 indicates an internally consistent data product that has been subjected to quality checks and data management procedures. Established calibration procedures were followed.

5. Data Acquisition Materials and Methods:

Study Site

Observations occurred at The Oak Ridge National Laboratory's SPRUCE (Spruce and Peatland Responses Under Changing Environments) project (Hanson et al., 2017) at S1 bog at the U.S. Forest Service's Marcell Experimental Forest, in Minnesota, USA (47.51, -93.45). SPRUCE is an ecosystem-scale study that warms the vegetation and peatland soil profile to a depth of 2 m within 12.8-m diameter open-top enclosures. The peat was heated first at – 2 m for a period of 1 year starting in summer 2014 (deep peat heating phase), then heating of the overlying air began in August 2015 starting the whole ecosystem warming phase of the experiment.

Methods

Sample Collection:

Cores were extracted from hollows in each of the 10 experimental plots (4, 6, 8, 10, 11, 13, 16, 17, 19, and 20) and two ambient plots (7 and 21) on 3 dates (2021-06-23, 2021-08-24, 2022-06-21). On each sampling date, cores extending to 200 cm below the hollow surface were collected using a knife to cut out low density surface peat to a depth of approximately 30cm, and below that using a Russian corer. Peat is then divided into 10 cm intervals from 0 and 50 cm and 25 cm intervals below 50 cm depth. Sub-samples were then frozen on dry ice in the field and shipped to ORNL where they were stored at -80°C or -20°C prior to processing. The 125-150 cm and 175-200 cm increments were excluded from qPCR analyses.

Sample Processing:

A subsample of ~0.25g of field-wet peat was taken from peat collected from each of the previously described plots and depths, and DNA was extracted using the Omega Bio-Tek Mag-Bind Environmental DNA 96 Kit (Omega Bio-Tek) according to the manufacturer's protocol. DNA was eluted in 100 µL of Elution Buffer (Omega Bio-Tek). DNA extracts were further processed through the DNA Clean and Concentrator Kit-25 (Zymo Research) according to the manufacturer's protocol to remove contaminants and to concentrate the DNA in a final volume of 50 µL. The concentration of extracted DNA for each sample was quantified using the Qubit dsDNA HS Assay Kit (ThermoFisher Scientific).

The same general approach outlined by Kluber et al. (2020) was then used to quantify gene copy number g-1 dry soil for bacteria, fungi, and archaea. Analyses were set up in clear 96-well plates. Each qPCR reaction consisted of 18µl of MasterMix—6 µl H₂O, 1µl of forward primer (at 10 µM), 1µl of reverse primer (at 10 µM), and 10µl of SYBR Green SuperMix (Bio-Rad Laboratories, Inc.)—combined with 2µl of a 1:10 dilution of DNA from each SPRUCE sample or known standard. Three technical replicates were conducted for each sample and standard. The primers used to quantify bacterial, archaeal, and fungal gene abundance are described in Table 1:

Table 1: DNA primer sequences used to quantify bacterial, fungal, and archaeal gene copy number.

Primer name	Sequence (5' → 3')	Reference
Bacterial F - EUB338	ACT CCT ACG GGA GGC AGC AG	Lane, 1991
Bacterial R - EUB518	ATT ACC GCG GCT GCT GG	Muyzer et al., 1993
Fungal F - nu-SSU-1196F	GGA AAC TCA CCA GGT CCA GA	Borneman & Hartin, 2000
Fungal R - nu-SSU-1536R	ATT GCA ATG CYC TAT CCC CA	
Archaeal F - rRNA 915F	AGG AAT TGG CGG GGG AGC AC	Colwell et al., 2008
Archaeal R -rRNA 1059R	GCC ATG CAC CWCCTC T	Yu et al., 2005

Genomic DNA from *Escherichia coli* K-12 MG1655 (bacteria), *Saccharomyces cerevisiae* S288C (fungi), and *Methanococcus maripaludis* C5 (archaea) was used to create a standard linear curve for each microbial Kingdom of interest. DNA from qPCR standards was diluted to 1:10, 1:100, 1:1,000, and 1:10,000 and used to generate a standard linear curve relating the log of the gene copy number to the measured threshold value (Ct). The standard curve was used to calculate gene copy number g-1 dry soil in SPRUCE samples based on sample Ct values measured by the CFX96 Real-Time System (Bio-Rad Laboratories, Inc.). Gene copy numbers of technical replicates were averaged (mean).

6. Related References:

Details on the SPRUCE experiment can be found in:

Hanson, P.J., J.S. Riggs, W.R. Nettles, J.R. Phillips, M.B. Krassovski, L.A. Hook, L. Gu, A.D. Richardson, D.M. Aubrecht, D.M. Ricciuto, J.M. Warren, and C. Barbier. 2017. Attaining whole-ecosystem warming using air and deep-soil heating methods with an elevated CO₂ atmosphere. *Biogeosciences* 14:861–883. <https://doi.org/10.5194/bg-14-861-2017>

Borneman, J., & R.J. Hartin. 2000. PCR primers that amplify fungal rRNA genes from environmental samples. *Applied and environmental microbiology*, 66(10), 4356–4360. <https://doi.org/10.1128/AEM.66.10.4356-4360.2000>.

Colwell, F. S., S. Boyd, M.E. Delwiche, D.W. Reed, T.J. Phelps, & D.T. Newby. 2008. Estimates of biogenic methane production rates in deep marine sediments at Hydrate Ridge, Cascadia margin. *Applied and environmental microbiology*, 74(11), 3444–3452. <https://doi.org/10.1128/AEM.02114-07>.

Griffiths, N.A., L.A. Hook and P.J. Hanson. 2016. SPRUCE S1 Bog and SPRUCE Experiment Location Survey Results, 2015. Carbon Dioxide Information Analysis Center, Oak Ridge National Laboratory, U.S. Department of Energy, Oak Ridge, Tennessee, U.S.A.
<http://dx.doi.org/10.3334/CDIAC/spruce.015>

Hanson, P.J., J.S. Riggs, W.R. Nettles, M.B. Krassovski, and L.A. Hook. 2016. SPRUCE Whole Ecosystems Warming (WEW) Environmental Data Beginning August 2015. Oak Ridge National Laboratory, TES SFA, U.S. Department of Energy, Oak Ridge, Tennessee, U.S.A.
<https://doi.org/10.3334/CDIAC/spruce.032>.

Kluber, L.A., E.R. Johnston, S.A. Allen, J.N. Hendershot, P.J. Hanson, & C.W. Schadt 2020. Constraints on microbial communities, decomposition and methane production in deep peat deposits. PLoS One, 15(2), p.e0223744. <https://doi.org/10.1371/journal.pone.0223744>.

Lane, D.J., 1991 16S/23S rRNA Sequencing. In: Stackebrandt, E. and M. Goodfellow, Eds., Nucleic Acid Techniques in Bacterial Systematics, John Wiley and Sons, Chichester, 115-175.

Muyzer, G., E.C. de Waal, & A.G. Uitterlinden. 1993. Profiling of complex microbial populations by denaturing gradient gel electrophoresis analysis of polymerase chain reaction-amplified genes coding for 16S rRNA. Applied and environmental microbiology, 59(3), 695–700. <https://doi.org/10.1128/aem.59.3.695-700.1993>.

Wilson, R.M., A.M. Hopple, M.M. Tfaily, S.D. Sebestyen, C.W. Schadt, L. Pfeifer-Meister, C. Medvedeff, K.J. McFarlane, J.E. Kostka, M. Kolton, & R.K. Kolka. 2016. Stability of peatland carbon to rising temperatures. Nature Communications, 7(1), p.13723.
<https://doi.org/10.1038/ncomms13723>.

Yu, Y., C. Lee, J. Kim, & S. Hwang. 2005. Group-specific primer and probe sets to detect methanogenic communities using quantitative real-time polymerase chain reaction. Biotechnol. Bioeng., 89: 670-679. <https://doi.org/10.1002/bit.20347>.

7. Data Access and Project Description:

For public access to SPRUCE data please visit the SPRUCE Web Site (<http://mnspruce.ornl.gov/datasets>), **ESS-DIVE** (<https://ess-dive.lbl.gov>), or **SPRUCE's data portal at ESS-DIVE** (https://data.ess-dive.lbl.gov/portals/SPRUCE_Experiment).

Contact for Data Access Information: Melanie Mayes - mayesma@ornl.gov

Description and Links to Supplemental Information:

- USFS Marcell Experimental Forest Website:
 - <https://www.nrs.fs.usda.gov/ef/marcell/>
- SPRUCE Project Website:
 - <https://www.ornl.gov/project/spruce-and-peatland-responses-under-changing-environments-spruce>

SPRUCE Project Description

The SPRUCE (Spruce and Peatland Responses Under Changing Environments) Experiment implemented 10 years of whole-ecosystem warming, deep peat heating, and elevated carbon dioxide (CO₂) treatments from 2015-2025. It was a primary component of the [Terrestrial Ecosystem Science Scientific Focus Area](#) at ORNL, which is focused on terrestrial ecosystems and the mechanisms that underlie their responses to environmental change. The experimental work was conducted in a *Picea mariana* [black spruce] – *Sphagnum* spp. bog forest in northern Minnesota, 40 km north of Grand Rapids, in the USDA Forest Service Marcell Experimental Forest (MEF). The site is located at the southern margin of the boreal peatland forest. It is an ecosystem considered especially vulnerable to environmental change, and anticipated to be near its tipping point with respect to future change. Responses to warming and interactions with increased atmospheric CO₂ concentration are anticipated to have important feedbacks on the atmosphere and carbon cycle, because of the high carbon stocks harbored by such ecosystems.

Experimental work in the 8.1-ha S1 bog is an environmental change manipulation focusing on the combined responses to multiple levels of warming at ambient or elevated CO₂ (eCO₂) levels. The experiment provides a platform for testing mechanisms controlling the vulnerability of organisms, biogeochemical processes and ecosystems to environmental change (e.g., thresholds for organism decline or mortality, limitations to regeneration, biogeochemical limitations to productivity, the cycling and release of CO₂ and CH₄ to the atmosphere).

The manipulation evaluates the response of the existing biological communities to a range of warming levels from ambient to +9°C, provided via large, modified open-top chambers. The +0, +2.25, +4.5, +6.75 and +9°C warming treatments were also conducted at eCO₂. The eCO₂ treatment was imposed as +500 ppm above ambient reference plots with actual concentrations ranging from 800 to 900 ppm. Both direct and indirect effects of these experimental perturbations will be analyzed to develop and refine models needed for full Earth system analyses.

Marcell Experimental Forest

Streamflow, weather, and well data collection began on the Marcell Experimental Forest in 1960. This 1100-ha site has six calibrated watersheds, each consisting of a mineral soil upland and organic soil peatland; an intermittent or perennial stream drains each peatland and its larger watershed. Formally established in 1962, the Marcell contains two units on land owned by the USDA Forest Service, Chippewa National Forest, State of Minnesota, Itasca County, and a private individual. Previous and ongoing research addresses the ecology and hydrology of peatland. Research concerns typical upland/wetland watersheds in the Lake States, atmospheric chemistry, nutrient cycling, soil quality, tree-stand dynamics, and a variety of watershed treatments applied to upland or bogs to investigate impacts on water yield, peak streamflow, water quality and nutrient processing.

SPRUCE Sponsor

Research sponsored by the [Office of Biological and Environmental Research](#) within the [U.S. Department of Energy's Office of Science](#).

The SPRUCE experiment is a multi-year cooperative interaction among scientists of the [Oak Ridge National Laboratory](#) operated by UT-Battelle, LLC and the U.S. Forest Service, [Northern Research Station](#), [Marcell Experimental Forest](#).

